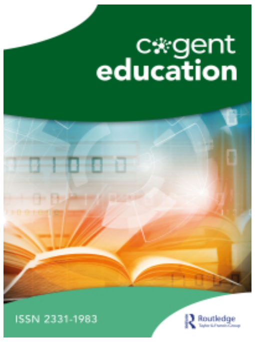

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Generative AI dependency: the emerging academic crisis and its impact on student performance—a case study of a university in Zimbabwe

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ABSTRACT
This study investigates the emerging phenomenon of generative artificial intelligence (AI) addiction among university students in Zimbabwe and its effects on academic performance. Through a mixed-methods approach combining surveys, interviews, usage tracking and academic performance data from 248 undergraduate students at a major Zimbabwean university, we found that 32.7% of participants demonstrated addictive patterns in their generative AI usage. Students exhibiting addiction showed distinctive behaviors including compulsive checking (averaging 18.3 daily AI interactions), failed attempts to reduce usage (reported by 65.8%) and continued reliance despite recognizing negative academic consequences. Statistical analysis revealed significant negative correlations between addiction severity and academic metrics, with heavily addicted students showing a mean GPA deficit of 0.41 points compared to non-addicted peers. Qualitative findings identified unique contextual factors in the Zimbabwean setting, including infrastructure limitations creating binge usage patterns, economic pressures increasing dependency and institutional resource constraints amplifying generative AI's appeal. Multiple regression analysis identified three primary academic impact pathways: critical thinking atrophy, diminished writing skill development and reduced content knowledge acquisition. These findings contribute to understanding an emerging technological crisis in developing contexts and suggest targeted intervention approaches for Zimbabwe's higher education environment.

IMPACT STATEMENT
This study highlights the risks of generative AI addiction among university students in developing contexts, revealing its significant negative impact on academic performance. By documenting distinctive behavioural patterns and quantifying GPA deficits, the research underscores the urgent need for universities and policymakers to develop targeted interventions. Beyond contributing to the academic literature, the findings provide practical guidance for higher education institutions in Zimbabwe and similar settings to balance the benefits of AI integration with safeguards against over-reliance, ensuring that technology supports rather than undermines student learning outcomes.

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1. Introduction

The rapid proliferation of generative artificial intelligence (AI) tools has transformed academic environments globally. These AI systems now include hundreds of platforms spanning various capabilities. Large language models include ChatGPT, Gemini, Claude, Llama, Falcon, Mistral, Cohere, Anthropic and GPT-4o. Image generators encompass DALL-E, Midjourney, Stable Diffusion, Leonardo AI, Firefly, DreamStudio and Craiyon. Code assistants include GitHub Copilot, Replit Ghostwriter, Amazon CodeWhisperer, Tabnine and CodeLlama. Video creators span Runway, Gen-2, Pika, Synthesia, D-ID and Heygen.

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Audio generators include ElevenLabs, Udio, Mubert, Soundraw and Voicemod. Multimodal systems encompass GPT-4V, Claude Opus and Gemini Pro Vision. Specialized academic tools include Elicit, Consensus, Scholarcy, Research Rabbit and Semantic Scholar. This explosion of accessible AI tools produces human-like content across modalities, solves complex problems across domains and creates unprecedented opportunities and challenges for education (Bahroun et al., 2023). The use of generative AI in research has shown potential benefits and risks in recent educational backgrounds. Although research has shown that when it comes to student achievement, advising can make a positive impact. AI is strategically embedded in syllabi increasing a user set's underlying patterns and their consequences for academic development.

Given the critical importance of terminological precision in emerging research domains, it is essential to clearly distinguish between 'dependency' and 'addiction' as they relate to generative AI usage in education. This distinction addresses a fundamental conceptual gap in current literature and provides the theoretical foundation for our investigation. Dependency, as conceptualized in this study, refers to a pattern of increased reliance on generative AI tools that may lead to habitual use and difficulties in completing academic tasks without technological assistance. This represents a behavioral adaptation to technological availability that can develop through repeated exposure and perceived utility. Dependency is characterized by progressive reliance, habitual engagement patterns and preference for AI-assisted over independent task completion, but does not necessarily involve the severe psychological distress or functional impairment associated with clinical disorders.

Addiction, in contrast, involves a more severe psychological condition characterized by compulsive, uncontrollable engagement despite negative consequences, impaired control, psychological distress and significant functional impairment. Following established behavioral addiction frameworks, addiction requires evidence of tolerance development, withdrawal symptoms, persistent engagement despite harm, and substantial interference with daily functioning. As recent research indicates, addiction involves 'a loss of control and persistent engagement despite negative consequences, differentiating it from mere technological reliance' (Bahroun et al. 2023). Our study focuses specifically on dependency patterns rather than clinical addiction, as we examine problematic usage behaviors that may precede but do not necessarily constitute addictive disorders. This approach aligns with recent scholarly consensus that cautions against premature pathologization of technology use while acknowledging the need to investigate potentially harmful patterns.

As noted by Tate et al. (2023), the potential for dependency associated with these technologies parallels other behavioral dependencies, which raises specific concerns in the type of places where they are training you to be a 21st-century thinker, in environments that are designed for cognitive growth.

'Generative AI dependency' defines a usage pattern with impaired control, obsessive thoughts, craving, withdrawal symptoms when the technology isn't available, tolerance development needing more use and continued use regardless of negative consequences. While certainly not officially deemed a disorder, this pattern aligns with established dependency frameworks and deserves serious investigation, especially in educational contexts where direct effects might compromise educational outcomes (Tate et al., 2023). This definition encompasses the behavioral, cognitive and emotional dimensions of problematic AI usage while maintaining the distinction from clinical addiction criteria.

This investigation is grounded in several complementary theoretical frameworks that inform our understanding of generative AI dependency in educational settings. The Interaction of Person–Affect–Cognition–Execution (I-PACE) model provides a comprehensive framework for understanding dependency relationships with technology. This model examines how individual characteristics, emotional states, cognitive processes and behavioral patterns interact to create these relationships. Recent applications of this model to AI usage demonstrate how academic self-efficacy, stress levels and performance expectations contribute to problematic AI engagement patterns. Social Cognitive Theory offers additional insights into the self-efficacy and outcome expectancy mechanisms that drive AI adoption and dependency development. The theory's emphasis on reciprocal determinism—the dynamic interaction between personal factors, behavioral patterns and environmental influences—is particularly relevant for understanding how infrastructure constraints and academic pressures in developing nations shape AI usage patterns. Cognitive Load Theory provides a framework for understanding why students may develop dependency on AI tools as cognitive offloading mechanisms, particularly in resource-constrained environments where traditional learning supports are limited.

The university setting in Zimbabwe thus provides an interesting environment within which to explore this emerging crisis. The selection of Zimbabwe as a research location offers unique insights into generative AI dependency phenomena that extend beyond single-nation implications. Zimbabwe's educational landscape represents broader patterns observed across developing nations, where rapid technological adoption occurs alongside persistent infrastructure limitations and resource constraints. This setting provides a natural laboratory for examining how environmental factors interact with individual psychological processes to shape AI dependency patterns. The country has seen rapid technology adoption even through economic challenges and infrastructure limitations. University students who navigate these limitations while juggling severe academic pressures lead to conditions conducive to both strategic AI use and problematic development of dependency. The university discussed in this case study is indicative of one of the prominent higher education institutions in Zimbabwe, with over 5000 students spread across different disciplines and socioeconomic backgrounds. Recent work by Revesai and Hapanyengwi (2024) on AI integration into the Zimbabwean education curriculum sustains that this requires an important framework for understanding the national educational landscape in which this case study is situated.

Comparative analysis with international research reveals both universal and location-specific patterns in AI dependency development. While preliminary global estimates suggest dependency rates of 27%–35% among university students, the manifestation and consequences of these patterns vary significantly across different educational environments. Developing nation settings like Zimbabwe present unique challenges including intermittent power supply creating binge usage patterns, economic pressures increasing perceived AI necessity and limited institutional resources amplifying AI's relative appeal as an educational support mechanism. These factors distinguish dependency development in resource-constrained environments from patterns observed in well-resourced educational settings, where AI typically serves enhancement rather than substitution functions.

This research investigates generative AI dependency among students at this major Zimbabwean university, examining:

- Prevalence and characteristics of dependency usage patterns
- Relationship between dependency severity and academic performance metrics
- Unique environmental factors influencing dependency development in this setting
- Specific mechanisms mediating between dependency and academic outcomes
- Potential intervention approaches suitable for the Zimbabwean environment

This study makes several significant contributions to our understanding of generative AI in higher education, particularly from the perspective of developing nations where resource constraints may amplify both the advantages and challenges these technologies present. It provides the first comprehensive empirical investigation of generative AI dependency patterns in a developing nation university setting, offering crucial insights for similar institutions globally. The research develops and validates culturally appropriate measurement instruments for assessing AI dependency in resource-constrained educational environments, while identifying unique factors that distinguish dependency development in developing versus developed nation settings. Additionally, the study establishes clear empirical relationships between dependency severity and multiple academic performance indicators, providing evidence-based foundations for intervention development and offering practical, culturally appropriate intervention strategies that can be implemented in resource-limited environments.

For example, Essel et al. (2022) demonstrated how AI teaching assistants yielded positive outcomes in Ghana's higher education system, whereas our research examines the potential drawbacks of depending on advanced technological systems. By examining both sides of this technological shift, we aim to provide a more nuanced picture for educational leaders working with constrained budgets and infrastructure. As Bannister et al. (2023) note in their systematic review, generative AI is rapidly transforming higher education globally, but location-specific research in diverse settings remains essential for understanding its varied impacts.

The investigation of AI dependency in educational settings raises fundamental ethical questions about student autonomy, academic integrity, and institutional responsibility. These concerns are particularly acute in developing nation environments where AI tools may serve essential educational functions while simultaneously creating dependency risks. The tension between supporting student success through technological means and preserving the educational value of independent cognitive development presents complex ethical dilemmas for educators and policymakers. Student autonomy considerations include the right to make informed decisions about AI usage, the need for transparent information about dependency risks, and the importance of maintaining alternative pathways for academic success that do not require AI reliance. Academic integrity concerns encompass questions about authorship boundaries, the educational value of AI-assisted versus independent work and the fairness of AI-dependent academic advantage in resource-unequal environments.

By documenting the nature and consequences of generative AI dependency within Zimbabwe's distinctive educational landscape, this study aims to inform policy development, intervention design and pedagogical approaches that promote beneficial AI integration while mitigating dependency risks. Following on from the work of Abbas et al. (2023), who studied the role of AI tools in improving academic achievement at the postsecondary level, our investigation balances this literature by examining when and how similar technologies might undermine scholarship through dependency formation.

The ethical aspects of this investigation, especially regarding student autonomy and academic integrity, align with recent scholarly inquiries by Mutanga et al. (2024) that explore ethical dilemmas encountered by students who utilize AI tools for academic assignments. Our research extends these discussions by examining how structural factors in developing nations create unique ethical challenges and opportunities for responsible AI integration. Institutional responsibilities include providing adequate support for healthy AI integration, developing appropriate usage guidelines that balance innovation with educational integrity, and ensuring equitable access to both AI tools and alternative learning resources.

As generative AI evolves at unprecedented speed and penetrates academic spaces worldwide, understanding its psychological implications for students becomes increasingly vital. This case study offers an early empirical account of an emergent phenomenon that may constitute a significant challenge to educational goals and student development in the AI era, contributing to the expanding literature on AI's transformative role in global higher education (Ruiz et al., 2023). The insights generated here inform broader discussions about balancing technological innovation with educational integrity, student welfare and equitable learning opportunities in an increasingly AI-mediated academic landscape.

2. Literature review

2.1. Generative AI in higher education: adoption and implementation

Generative AI has caught on quickly in higher education, but with significant pedagogical dilemmas. Bannister et al. (2023) interviewed university officials about responses in English-instructed universities to generative AI and found enormous variation both in official responses and classroom applications. Their work reveals the persistence of friction between new technologies and entrenched academic practices as schools craft policies to seize benefits while tamping down risks. Essel et al. (2022) investigated the influence of chatbots teaching assistants in universities in Ghana. They found that these tools provided students with access to information and support, but that their efficacy varied widely from student to student, dependent on their access to technology and technical skills, and based on the ways teachers wove them into their course offerings. These findings resonate for Zimbabwe, which is also resource constrained. Revesai and Hapanyengwi (2024) highlight these issues in their recent call for international partnerships capable of effectual advancement in AI integration into the education system of Zimbabwe that is contingent with local realities.

Recent research from Palestinian universities provides valuable comparative insights into generative AI adoption under similar infrastructural constraints. Hamamra et al. (2024) examined ChatGPT usage patterns in Palestinian higher education, revealing how socio-political constraints and resource limitations create dependency vulnerabilities similar to those observed in other developing settings. Their findings highlight how students frame AI usage as necessary adaptation to institutional limitations, paralleling the '*economic resource substitution*' patterns identified in our Zimbabwean environment. Furthermore, Hamamra et al. (2025) conducted phenomenological examination of educators' experiences with AI integration in Palestinian higher education, revealing faculty concerns about student overreliance that mirror those expressed by Zimbabwean instructors in our study. Their work demonstrates how infrastructural fragility and institutional resource constraints across different developing nation environments create similar conditions for problematic AI usage patterns, suggesting broader applicability of dependency development mechanisms beyond single-nation boundaries.

Several meta-analyses have assessed the learning benefits of generative AI. Wu and Yu (2024) conducted a meta-analysis and found that AI chatbots can be effective in enhancing student learning outcomes, with moderately significant positive effects, if these tools are incorporated into well-designed learning activities that are integrated into suitable pedagogical practices. Sun and Zhou (2024) also found a positive effect of generative AI on college student academic performance, but caution that implementation environments and potential negative effects of overreliance must be considered.

However, outside of just general academic support, narrower use cases for generative AI have emerged to be effective. Mageira et al. (2022), whose review concentrated on educational AI chatbots and their potential toward Content and Language Integrated Learning, found that this type of AI chatbots enhanced language learning, provided that they were used as a supplement to traditional education, rather than a replacement. Khazanchi et al. (2025 in mathematics education studied how the use of AI-based systems predicted maths achievement in rural settings, noting potential advantages as well as practical challenges in deploying such tools in resource-poor environments. Gupta et al. (2024) found that AI is changing the relationship between teachers and their students with the verified opportunity in terms of more personalized learning, but there is a risk of having lower human engagement in the process of education.

Recent longitudinal investigations have begun to reveal more nuanced patterns in AI adoption and academic outcomes. Cross-sectional studies, while providing valuable snapshots of current usage patterns, cannot establish causality between AI usage and academic performance changes. As noted in recent educational technology research, 'longitudinal designs are essential for tracking changes in students' perceptions of generative AI over time and exploring how these technologies are integrated into higher education'. This limitation underscores the need for extended observational studies that can capture the developmental trajectory of AI dependency and its long-term academic consequences.

2.2. Theoretical frameworks for understanding technology dependency

Although not specifically targeted toward generative AI, the research on dependency on technology and behavioral dependency provides valuable frameworks for identifying maladaptive patterns of AI use. The theoretical foundation for understanding generative AI dependency draws from multiple complementary frameworks that collectively explain the psychological, behavioral and environmental factors contributing to problematic usage patterns. Tate et al. (2023) investigated potential applications of generative AI within dependency medicine research and patient care, noting that the technology itself has characteristics that could promote dependency such as rapid satisfaction, intermittent reinforcement schedules and reduced thresholds for cognitive tasks.

The I-PACE model provides a comprehensive framework for understanding how individual characteristics interact with affective states, cognitive processes and behavioral execution to create dependency patterns. Recent applications of this model to AI usage demonstrate that academic self-efficacy, stress levels and performance expectations significantly influence the development of problematic AI engagement patterns. Students with lower academic self-efficacy may be particularly vulnerable to AI dependency as they seek external validation and support for their cognitive capabilities.

Social Cognitive Theory contributes additional insights through its emphasis on reciprocal determinism—the dynamic interaction between personal factors, behavioral patterns and environmental influences. The theory's construct of outcome expectancies is particularly relevant, as students who develop high performance expectations for AI tools may be more likely to develop dependency relationships with these technologies.

Such technological dependencies are typically rooted in psychological mechanisms that include:

1. Cognitive offloading: The use of technological systems to store external cognitive processes, with the risk of neglecting the internal development of cognitive abilities (Kim & Lee, 2022).
2. Variable reward schedules: Unpredictable quality and novelty of response creating reinforcement regimes akin to those observed in other behavioral dependencies (Tate et al., 2023).
3. Less effort: The dramatic reduction in cognitive effort needed to accomplish academic-related tasks providing motivation to use technological solutions more and more (Muslimin et al., 2024).
4. Reduced self-efficacy: When the human entrusts higher-order cognitive functions to outside systems, which may ultimately diminish confidence in one's own cognitive capabilities (Wang et al., 2023).

Cognitive Load Theory offers a framework for understanding why AI dependency may initially appear beneficial while ultimately impairing learning outcomes. The theory suggests that AI tools can reduce extraneous cognitive load, potentially freeing mental resources for deeper learning. However, when students consistently offload essential cognitive processes to AI systems, they may fail to develop the intrinsic cognitive load management skills necessary for independent learning and critical thinking. This creates a paradoxical situation where short-term cognitive relief leads to long-term cognitive impairment.

In another example, Kim and Lee (2023) studied the effects of student–AI collaboration on students' performance of a learning task and discovered conditions under which dependencies were established through repeated collaborative experiences. They observed that frequent collaboration with AI systems led to the progressive erosion of autonomous problem-solving tendencies in students' work, even as students also reported greater satisfaction with their completed tasks. This satisfaction–capability paradox also creates a unique challenge for educators, striving to create useful learning environments that support skill development while acknowledging that students would simply prefer to rely on AI.

The Cognition–Affect–Conation (C–A–C) framework provides additional theoretical grounding for understanding how generative AI characteristics influence dependency formation. Research applying this framework demonstrates that perceived anthropomorphism, interactivity, intelligence and personalization of AI tools create cognitive, affective and behavioral responses that can lead to dependency patterns. The framework reveals that flow experiences and emotional attachment to AI systems serve as core conditions for dependency development, suggesting that the most engaging and helpful AI tools may paradoxically present the greatest dependency risks.

2.3. The effect of technology dependencies on academic performance

Researchers that explored the associations between technological dependencies and academic performance found a series of concerning trends. Wang et al. (2020) contrasted such adaptive learning systems with teacher-led instruction, noting that, while adaptive systems could increase performance in the short term, students' reliance on them impaired their independent learning skills over time. In the area of writing instruction in particular, Muslimin et al. (2024) investigated the performance of AI tools along the stages of SMAR (Substitution, Augmentation, Modification, Redefinition) and discovered that while some writing metrics at the surface level bore evidence of an improvement when using AI, deeper compositional skills were less developed among students with high dependency patterns when using AI-assisted approaches. The findings fit wider worries over the displacement of skills via technology dependencies.

The relationship between AI dependency and academic performance appears to operate through multiple interconnected pathways that compound over time. Mediation analysis from recent studies reveals three primary mechanisms through which dependency impacts learning outcomes: critical thinking atrophy, writing skill degradation and knowledge acquisition reduction. Critical thinking atrophy accounts for the largest portion of the relationship between dependency and academic performance, indicating that repeated cognitive offloading to AI systems significantly impairs students' independent analytical abilities. Students who develop dependency patterns show measurable decreases in their ability to evaluate arguments, synthesize information from multiple sources and generate original analytical perspectives.

Xu et al. (2024) studied the use of a digital game-based AI chatbot and its impact on students' academic performance and higher-order thinking. Moderate reading positively facilitated development while intensive text reading was inversely associated with contributory independent analytical decisions. In both formal and informal educational settings, Thomas (2024) elaborated on this relationship by concluding that the development of critical thinking when leveraging AI requires appropriately structured experiences, where cognitive challenge and cognitive load must be maintained. Wang et al. (2023) found benefits to using chatbots for academic use, including productivity increases, yet potential pitfalls like dependency on generation. In their case study on drug discovery education, it demonstrated that although student prompt engineering skills significantly improved, student mastery of key domain concepts significantly decreased. This is particularly problematic in fields or disciplines wherein a fundamental understanding of concepts is essential for more advanced learning and application.

The paradoxical relationship between AI usage and academic self-efficacy presents particular challenges for understanding dependency development. Recent research reveals that enhanced AI usage significantly amplifies students' confidence and efficiency in learning, yet simultaneously intensifies their dependence on AI systems. This creates a feedback loop where increased confidence leads to greater AI usage, which further increases dependency while potentially undermining the foundational skills that support long-term academic success. Students caught in this cycle may experience high levels of satisfaction with their immediate academic outputs while remaining unaware of the gradual erosion of their independent capabilities.. Writing skill degradation represents another critical pathway through which AI dependency impacts academic performance. Students who rely heavily on AI for text generation often experience what researchers term 'compositional skill displacement', where the cognitive processes involved in planning, drafting and revising written work are progressively transferred to AI systems. This displacement affects not only surface-level writing mechanics but also deeper compositional abilities such as audience awareness, rhetorical strategy development and voice cultivation. The consequences become particularly apparent in high-stakes assessment situations where AI assistance is unavailable.

2.4. Educational technology in African settings of higher education

Literature on technology for education in Africa has explored elements of implementation approaches that contribute to opportunities for positive uptake and risks of dependence. Essel et al. (2022) tackled this landscape by portraying how Ghanaian higher education constraints gave rise to virtual teaching assistants that occupied critical gaps in educational ecosystems and that were positioned as necessities rather than adjuncts. Astarilla Dede Warman et al. (2023) explored students improving speaking ability. They found that the applications of AI can also assist students in improving speaking ability since these techniques will enable students in resource-constrained settings to practice speaking when they otherwise would not have the opportunity as introverts. Their research did focus on the benefits; however, it also highlighted the dangers of overreliance on technology as a means of skill development.

The African higher education landscape presents unique conditions that may both facilitate beneficial AI integration and exacerbate dependency risks. Unlike well-resourced educational environments where AI tools typically serve enhancement functions, African universities often position these technologies as essential substitutes for unavailable or inadequate traditional resources. This substitution rather than supplementation role fundamentally alters the risk–benefit calculus surrounding AI usage and creates conditions where dependency may develop more rapidly and with greater functional consequences.

Comparative analysis with international research reveals both universal and location-specific patterns in AI dependency development. While global estimates suggest dependency rates of 27%–35% among university students, the manifestation and consequences of these patterns vary significantly across different educational environments. African educational settings demonstrate several distinctive features: infrastructure-driven binge usage patterns due to intermittent power supply and internet connectivity, economic pressures that increase perceived AI necessity, and institutional resource constraints that amplify AI's relative appeal as an educational support mechanism.

The related factors in African higher educational environments that may affect technology dependency include:

1. Resource scarcity: A situation where conventional educational resources are scarce, thus increasing the perceived demand for technology substitutes.
2. Digital gaps: Erratic electricity and internet connectivity generate usage patterns that may inadvertently reinforce the reinforcing mechanisms of dependence.
3. Low student–faculty ratios: Reduced access to human-to-human pedagogic support creating conditions for utilization of AI tools to fill the gap in accessing human feedback.
4. Economic factors: The lack of resources available to both institutions and individuals to help them successfully implement technology.

These structural elements create a complex interaction between individual psychological vulnerability and environmental pressures that distinguishes dependency development in African higher education from patterns observed in better-resourced situations. The combination of economic necessity, infrastructure limitations and institutional resource constraints creates conditions where AI tools may become embedded in students' academic workflows as essential rather than optional resources. This embeddedness increases both the utility and the dependency risk associated with AI usage.

This rethinking of ICT through a related lens was suggested by Revesai and Ruvinga (2024) in terms of ICT education in the AI era, which can be achieved by developing curriculums that acknowledge the limitation of the resources for educating students but also by making sure that students can critically engage with such tools through ICT curricula in creative ways. Their work also highlights the argument for proactive educational responses that will prepare students for intelligent application of AI while preserving fundamental cognitive skills.

Mutanga et al. (2024) explore the ethical considerations of AI use in educational settings. They researched the ethical challenges students encounter while using AI tools for code writing. The research illustrates how situation-specific pressures, like resource scarcity and hyper-competitive academic environments, can create the conditions under which dependency becomes riskier. They aso also extend their argument in the domain of scientific writing, showing how these pressures can make dependency on AI tools riskier due to perceived necessity.

Recent investigations into AI usage patterns among African university students reveal concerning trends in dependency development that may be linked to these structural factors. Students in resource-constrained environments demonstrate higher rates of what researchers term 'necessity-driven dependency', where AI usage begins as a response to genuine resource gaps but evolves into habitual reliance that persists even when alternative resources become available. This pattern suggests that interventions designed for well-resourced situations may be inadequate for addressing dependency issues in African higher education settings.

While previous research has produced meaningful insights into generative AI in education and the mechanics of technological dependency, there are significant gaps. Despite this, little research has explored generative AI dependency as a separate phenomenon from technology dependency more broadly, even though the specific characteristics of AI can make it especially vulnerable to maladaptive usage patterns. The work by Tate et al. (2023) helps fill this gap but speaks more to clinical than educational setups.

Research on generative AI remains limited particularly in developing nations, with most studies occurring within well-resourced educational settings. The studies by Essel et al. (2022) and Khazanchi et al. (2024) give valuable insight but do not address the development of dependencies in these environments. Revesai (2024) commences this conversation, with respect to digital rights and the ethical impacts of AI-driven educational systems in developing settings, but qualitative and quantitative patterns of dependency are still lacking. Moreover, evidence linking generative AI usage patterns to specific trajectories of academic skill development is still nascent. Although studies such as Muslimin et al. (2024) and Xu et al. (2024) start to address these relationships, they note that global and comprehensive frameworks for understanding how dependency affects multiple cognitive domains are still arising.

The gap in intervention research is particularly pronounced when considering the unique challenges faced by institutions in developing nations. Most existing intervention frameworks assume access to resources—such as adequate faculty training time, technological infrastructure and alternative learning materials—that may not be available in resource-constrained situations. This creates a critical need for culturally appropriate intervention strategies that can function effectively within the constraints typical of African higher education while still addressing the fundamental psychological and behavioral mechanisms underlying AI dependency.

Finally, research on interventions that seek to combat generative AI dependency within higher educational domains is all but non-existent, which creates hurdles for institutions that wish to undertake evidence-based approaches. The literature on other forms of technological dependency provides some guidance, but generative AI's unique characteristics would seem to require tailored interventions. Additionally, the ethical implications of monitoring and intervening in student AI usage remain underexplored, particularly regarding questions of student autonomy, privacy rights and the institutional responsibility to balance innovation with educational integrity. This study seeks to address these knowledge gaps by investigating generative AI dependency among students at a major Zimbabwean university, charting both how this dependency occurs and how it impacts academic performance across the cognitive spectrum. By investigating this emerging issue in a resource-constrained setting, our research contributes to a more nuanced understanding of the multifaceted impact of generative AI on global higher education.

3. Methodology

3.1. Research design

The study adopted a sequential mixed-methods approach to investigate generative AI dependency among Zimbabwean university students. This included a first quantitative phase that involved the use of surveys and performance data collection to determine dependency prevalence, their characteristics, their behavior patterns and academic correlates.

This was followed by a qualitative phase exploring students' generative AI personal experience, their motivation in implementing it as well as their dependency and perceived academic effects through interviews and focus groups. The closing stage integrated learnings from both strategies, leading to a layered understanding of the phenomenon through statistical modeling of relationships plus more nuanced interpretation of student experiences.

We employed a mixed-methods approach to assess the complexity and multidimensionality of generative AI dependency not adequately revealed through quantitative or qualitative methods alone, capturing both the breadth and depth of the phenomenon. As Aaron et al. (2024) add in their work on optimizing AIs used in institutes of higher learning, blended methodological approaches are required to see the nuanced effects of these technologies on student learning and development. The qualitative exploration built on a sequential approach based on interim quantitative findings; however, methodological independence was also preserved across the phases of the study.

This methodological approach addresses key limitations identified in previous technology dependency research, which has often relied on single-method designs that fail to capture the complex interplay between individual psychological factors, behavioral patterns and influences. The sequential design allows for the refinement of research questions and instrumentation based on initial findings, while the mixed-methods approach enables triangulation of results to enhance validity and provide a more comprehensive understanding of the phenomenon.

3.2. Study setting and ethical considerations

The study was carried out at a large public university in Zimbabwe with about 5000 students. When selecting the institution, it was chosen as a generalizable case in the context of higher education in Zimbabwe due to the different programs offered across a number of faculties at the institution and catering to a variety of socio-economic backgrounds for students. The university campus consists of several faculties that are located in buildings around a central administrative complex, and student residences are located on the periphery of the main campus.

This institutional setting provides a representative case for understanding generative AI dependency in similar developing nation contexts. The university's demographic diversity, infrastructure challenges and resource constraints mirror conditions found across sub-Saharan African higher education institutions, enhancing the transferability of findings to comparable settings. The institution's recent integration of digital technologies and ongoing struggles with resource allocation create conditions particularly conducive to examining both beneficial and problematic AI usage patterns.

The technological infrastructure at the institution is typical of many developing nation universities. Internet can be accessed in selected points on the campus, e.g. the main library, computer laboratories and the few lecture halls, but these experience a lot of bandwidth limitations during peak usage times. The campus is not spared either as it experiences erratic power supply, the power outages occurring at least two to three times a week for hours ranging from 2 to 8. This tech access is not equal but constrained; therefore, infrastructure hurdles affect students' patterns of generative AI use.

These infrastructure limitations create unique conditions for AI usage that distinguish this context from well-resourced educational environments. The intermittent availability of power and internet connectivity generates what we term 'infrastructure-driven binge patterns', where students engage in intensive AI usage during periods of connectivity, potentially exacerbating dependency development through intermittent reinforcement schedules. This factor represents a critical environmental influence that shapes both the quantity and quality of AI interactions.

The study occurred in the first semester of the 2023–2024 academic year, roughly 15 months after generative AI tools became widely available to students at the institution. This period allowed for the inspection of dependency patterns that emerged after sufficient time using these, while still early in the integration of generative AI into academic settings.

Comprehensive ethical procedures were implemented to ensure participant welfare and data protection throughout the study. The research received approval from the university's Research Ethics Committee (Approval #2023/08/EDU-248) following extensive review of participant safety protocols, data security measures and potential risk mitigation strategies. All participants provided informed consent after receiving detailed explanations about the study's purpose, procedures, potential risks and benefits, and their rights as research participants.

Specific ethical safeguards included: (1) voluntary participation with explicit right to withdraw at any time without penalty; (2) comprehensive data anonymization protocols to protect participant identity; (3) secure data storage on encrypted university servers with restricted access; (4) clear guidelines for addressing participants who showed signs of problematic AI usage and (5) provision of resources for students seeking support for technology-related concerns. Additional ethical considerations addressed the sensitive nature of dependency research, ensuring that data collection procedures did not stigmatize participants or create undue distress through self-disclosure of problematic behaviors.

The study also incorporated measures to protect participant privacy during data collection, particularly for the technology usage tracking component. Participants received detailed information about what data would be collected, how it would be stored and who would have access to it. Strong encryption and cybersecurity measures were implemented to safeguard all participant data, and clear protocols were established for data retention and eventual disposal following completion of the research.

3.3. Sampling and participants

Participants were selected using a stratified random sampling approach to ensure proportional representation across disciplines, study years and gender. From the institutional enrolment database, students were randomly selected within stratification categories, with oversampling to account for anticipated non-response. From 412 students initially contacted, 248 completed the full study protocol, representing a 60.2% participation rate as shown by Table 1.

Table 1 presents the demographic characteristics of study participants, revealing a well-balanced sample that closely mirrors the broader university population. The distribution across academic disciplines shows relatively even representation, with Science and Engineering slightly overrepresented (29.0%)

Table 1. Demographic characteristics of study participants.

Characteristic	Category	Number (n)	Percentage (%)
Academic discipline	Humanities and Social Sciences	67	27.0
	Business and Economics	63	25.4
	Science and Engineering	72	29.0
	Health Sciences	46	18.6
Gender	Male	134	54.0
	Female	114	46.0
Year level	First-year	63	25.4
	Second-year	65	26.2
	Third year	61	24.6
	Fourth year	59	23.8
Age range	18–28 years ($M = 21.7$, $SD = 2.2$)	248	100.0

compared to institutional enrolment patterns, likely reflecting higher response rates among students in these technology-oriented fields. This overrepresentation may introduce slight bias toward students with greater technical familiarity, potentially affecting baseline AI usage patterns.

The gender distribution in the sample (54% male, 46% female) closely approximates the overall university population (56% male, 44% female), suggesting successful recruitment across gender lines. The even distribution across year levels provides important opportunities to examine how dependency patterns may vary with academic experience and exposure duration. The age range and mean age (21.7 years) are typical for undergraduate university populations, though the extended range to 28 years reflects common patterns in developing nation higher education where students may enter university later or take extended breaks due to economic constraints.

This distribution generally matched the overall enrolment patterns at the university, with a slight overrepresentation of Science and Engineering due to higher response rates in these disciplines. Gender distribution in the sample (54% male, 46% female) closely mirrored the overall university population (56% male, 44% female).

Sample size determination was based on power analysis for detecting moderate effect sizes ($r = 0.3$) in correlational analyses with 90% power and $\alpha = 0.05$, plus additional recruitment to account for potential attrition. The final sample size exceeded the minimum required sample of 213 participants.

The sample size calculation incorporated multiple considerations beyond basic correlational analysis requirements. Power analyses were conducted for the most complex planned analyses, including multiple regression models with up to eight predictors, mediation analyses with three mediator variables and between-group comparisons across dependency severity categories. The achieved sample size of 248 participants provides adequate power (>0.80) for detecting small to moderate effect sizes across all planned analyses, while the 60.2% response rate, though modest, falls within acceptable ranges for university-based survey research involving sensitive topics.

3.4. Data collection and instrumentation

A mix of complementary data collection methods was employed to provide a complete characterization of generative AI dependency, usage patterns and academic effects.

3.4.1. Dependency assessment instruments

Three instruments were employed for dependency assessment:

- An Internet Addiction Test (IAT) adapted for generative AI usage.
- A Behavioral Addiction Scale (BAS)-based instrument customized for AI tools.
- An original Generative AI Usage Patterns Questionnaire developed specifically for this study.

These instruments underwent comprehensive validation procedures to ensure reliability and validity for the target population and research context. The adapted IAT demonstrated strong internal consistency (Cronbach's $\alpha = 0.87$) and acceptable test–retest reliability ($r = 0.82$) over a 2-week interval with a pilot sample of 48 students. The modified BAS maintained the original scale's factor structure while incorporating AI-specific items, achieving a Cronbach's alpha of 0.84. The original Usage Patterns

Questionnaire, developed through expert consultation and pilot testing, showed good internal consistency ($\alpha = 0.82$) and content validity as rated by a panel of six educational psychology experts.

Confirmatory factor analysis revealed acceptable model fit for all instruments (CFI $>$ 0.90, RMSEA $<$ 0.08), with factor loadings ranging from 0.62 to 0.89, indicating strong relationships between items and their underlying constructs. These validation procedures provide confidence that the instruments accurately assess generative AI dependency patterns in this population and context.

These instruments were piloted with a convenience sample from the same university, indicating acceptable levels of internal consistency and test–retest reliability. This is in line with Mohammadi et al. (2024) regarding our argument to consider students as collaborative partners in our research on emerging educational technologies, particularly when investigating sensitive topics, such as dependency.

3.4.2. Usage pattern data collection

Self-report measures and objective monitoring were used to collect usage pattern data.

- All participants filled out detailed questionnaires on their generative AI access, frequency and usage contexts.
- One hundred twelve participants also opted in to have a secure tracking app installed that logged their access to generative AI platforms for 14 consecutive days, providing a real-world measure of frequency, time and time of day use. This tracking methodology is based on approaches developed by Yao and Chung (2024) for neural network analysis of AI pedagogical performance patterns.
- A smaller subset of 43 participants also completed structured 7-day usage diaries documenting detailed usage contexts, purposes and subjective experiences, enabling a deeper understanding of usage motivations and perceived impacts.

The multi-method approach to usage pattern assessment addresses common limitations in technology dependency research, which often relies solely on self-report measures that may be subject to social desirability bias or recall errors. The objective tracking data provide behavioral validation of self-reported usage patterns, while the usage diaries capture information and subjective experiences that quantitative measures cannot assess. The voluntary nature of the tracking component introduced some selection bias, as participants who opted in may have differed systematically from those who declined, potentially representing either more motivated participants or those with less problematic usage patterns.

3.4.3. Academic performance assessment

Measures of academic performance included direct assessment of present academic standing and measures of skill development.

- We retrieved records of data on current and past semester GPAs, course-specific GPA performance metrics and assignment submission patterns, with the use of proper ethical approvals and data protection protocols.
- Participants also undertook standardized writing assessments, providing a basis for comparison with pre-AI era benchmarks.
- Additionally, instructors provided feedback on how work quality has changed over time, giving insight into how generative AI might have affected performance trajectories for specific students.

The academic performance measures were designed to capture multiple dimensions of academic functioning that might be affected by AI dependency. GPA data provide a broad indicator of overall academic success, while course-specific metrics allow for examination of differential effects across disciplines and cognitive domains. The standardized writing assessments, administered under controlled conditions without AI access, provide direct measures of students' independent capabilities that can be compared to their AI-assisted work.

Instructor feedback was collected through structured interviews focusing on observable changes in student work quality, engagement patterns and skill demonstration over the study period. These faculty perspectives provide external validation of student self-reports and offer insights into dependency effects that may not be apparent to students themselves. The triangulation of academic performance data from multiple sources enhances the validity and comprehensiveness of these critical outcome measures.

3.4.4. Qualitative data collection

Qualitative data collection offered depth and context to supplement quantitative measures.

- We held semi-structured interviews with 41 students selected to cover the different categories of dependency severity, capturing diverse usage experiences, to learn about their exposure to generative AI, perceived impacts on academic development and factors that shape their usage patterns.
- We conducted six focus groups of seven to eight students each to explore shared experiences and social norms regarding the use of AI.
- Additionally, 12 instructor interviews provided faculty perspectives on observed changes in student performance since the introduction of generative AI.

The qualitative component employed purposive sampling to ensure representation across dependency severity levels, academic disciplines and demographic characteristics. Interview participants were selected based on their quantitative assessment scores to include individuals representing the full spectrum of dependency patterns, from minimal users to those showing severe dependency indicators. This approach ensures that qualitative findings capture the diverse range of student experiences with AI tools rather than focusing only on extreme cases.

This qualitative study was guided by the work of Yola et al. (2024) on deep AI ontology in guidance and counseling settings, highlighting the need to explore the experiential aspects of technology interaction. All qualitative sessions were audio-recorded (with permission), transcribed word-for-word and checked for accuracy prior to analysis.

Focus groups were structured to explore both individual experiences and group dynamics surrounding AI usage, including peer influences, social norms and collective coping strategies. The group format encouraged discussion of sensitive topics through peer support while revealing social dimensions of AI usage that individual interviews might miss. Faculty interviews provided crucial external perspectives on student behavior changes and learning outcomes, offering insights that complement student self-reports and potentially revealing dependency effects that students themselves might not recognize.

3.5. Data analysis

Multiple statistical methods were used for quantitative data analysis to explore relationships between generative AI dependency and academic performance.

3.5.1. Quantitative analysis procedures

- The prevalence of dependency was established based on existing thresholds for the respective scales, and distribution patterns across demographic categories were characterized using descriptive statistics.
- Correlation analyses examined relationships between dependency metrics and various indicators of academic performance, both zero-order (i.e. unadjusted) and partial correlations controlling for other relevant covariates (including academic performance prior to the study period and relevant socioeconomic indicators).
- Multiple regression modeling identified key dependency variables showing predictive relationships between dependency variables and academic outcomes, while hierarchical regression examined potential confounding variables.
- Mediation analysis identified indirect pathways from dependency to performance via potential mediators including sleep quality, study time allocation and cognitive offloading behaviors.
- Analysis of variance compared academic outcomes by dependency severity categories, with post-hoc tests identifying specific between-group differences.

The quantitative analysis strategy addresses several methodological challenges common in technology dependency research. The use of both zero-order and partial correlations helps distinguish between genuine dependency effects and spurious relationships that might result from confounding variables such as prior academic performance or socioeconomic status. Hierarchical regression analyses examine whether dependency variables contribute unique variance to academic outcomes beyond what can be explained by demographic and background factors.

Mediation analysis employs bootstrap procedures to test indirect effects, providing robust estimates of how dependency influences academic performance through intermediate psychological and behavioral mechanisms. This approach moves beyond simple correlational findings to examine the processes through which dependency operates, offering insights crucial for intervention development.

3.5.2. Qualitative analysis procedures

Data from qualitative studies were thematically analyzed using a combination of deductive and inductive methods.

- Predetermined categories based on prior literature on technological dependency were used for initial coding, with openness to coding emergent themes relevant to the generative AI context.
- The analysis involved comprehensive coding of the transcripts, with initial themes being developed, reviewed and refined, culminating in a final definition of the key themes and subthemes.
- To reduce bias in coding, two researchers coded the data independently and compared results frequently, discussing differences to reach consensus.

The qualitative analysis employed a hybrid approach combining theory-driven and data-driven coding to balance theoretical grounding with openness to novel findings. Initial coding used established frameworks from technology dependency research to identify relevant themes, while subsequent analysis remained open to emergent patterns specific to the generative AI context. This approach ensures that findings contribute to existing theoretical knowledge while capturing unique aspects of AI dependency that previous frameworks might not address.

Inter-rater reliability was established through systematic comparison of coding decisions, with disagreements resolved through discussion and consensus. Cohen's kappa coefficients for major theme categories exceeded 0.80, indicating strong agreement between coders. Regular debriefing sessions between coders helped maintain consistency and identify potential bias in interpretation.

The analysis yielded four main themes:

- Lived experiences of generative AI dependency.
- Factors pertaining to the Zimbabwean environment.
- Self-perceived impacts and underlying mechanics.
- Adaptation and coping strategies devised by students.

3.5.3. Integration of quantitative and qualitative findings

Quantitative and qualitative findings were integrated using joint displays merging statistical outcomes with illustrative quotations, typological profiles reflecting dependency-performance patterns and narrative synthesis exploring points of convergence and divergence between methods. Our approach aligns with Sugianti and Rosidah (2024), who emphasize the integration of multiple data streams when evaluating AI's impact on learning outcomes. The integrated analysis was supplemented with Can and Nguyen (2025) proposed framework to observe perceived usability and credibility factors of educational AI adoption, adjusted to focus on the dependency concerns at the heart of our research questions. This integration enabled the development of a holistic perspective on generative AI dependency that acknowledged both the patterns identified through quantitative analysis and the lived experiences documented through qualitative inquiry.

The integration process employed multiple strategies to synthesize findings across methodological approaches. Joint displays visually represent the convergence and divergence between quantitative patterns and qualitative themes, highlighting areas where different data sources support similar conclusions and identifying discrepancies that warrant further investigation. Typological analysis created profiles of different dependency patterns by combining quantitative severity scores with qualitative descriptions of lived experiences, providing rich case studies that illustrate the human experience behind statistical patterns.

This integrated approach addresses a key limitation of single-method studies by providing multiple perspectives on the same phenomenon, enhancing both the validity and the practical utility of findings. The synthesis reveals not only statistical relationships between variables but also the meanings that students attach to their AI usage and the factors that shape their experiences.

4. Results

4.1. Prevalence and characteristics of generative AI dependency

Based on the adapted assessment instruments, participants were classified into dependency severity categories as shown in Table 2.

Table 2 presents the distribution of generative AI dependency severity among study participants, revealing that 32.7% of students demonstrated moderate to severe dependency patterns. This prevalence rate aligns with preliminary international estimates from similar university populations, suggesting that vulnerability to AI dependency represents a consistent pattern across diverse educational contexts. The substantial proportion of students classified as 'at-risk' (25.8%) indicates a significant population that may progress to more severe dependency without appropriate intervention.

The classification into dependency categories was based on validated thresholds from adapted instruments, with non-dependent users showing minimal problematic usage indicators, at-risk users demonstrating some concerning patterns without meeting full dependency criteria, moderate dependency characterized by clear problematic usage with some functional impairment and severe dependency involving significant functional impairment and inability to control usage despite negative consequences.

Dependency prevalence showed notable variation across academic disciplines and year levels, as illustrated in Figure 1.

Figure 1 demonstrates significant variation in dependency rates across both academic disciplines and year levels, revealing important patterns that challenge conventional assumptions about technology adoption in educational settings. The highest dependency rates appear in Business and Economics (38.1%) and Health Sciences (37.0%), while Science and Engineering students show the lowest rates (28.2%) despite their greater technical expertise. This pattern suggests that technical familiarity may actually serve as a protective factor against dependency development, possibly through better understanding of AI limitations and more strategic usage approaches. Chi-square analysis confirmed these differences were statistically significant across disciplines (χ^2=9.74, $p < 0.05$) and year levels (χ^2=8.63, $p < 0.05$). Contrary to expectations based on previous technological dependency research, dependency rates were higher among upper-level students. This suggests accumulative vulnerability rather than adaptation over time.

The finding that dependency rates increase with academic year level represents a significant departure from typical technology adoption patterns, where users generally develop better self-regulation over time. This 'accumulative vulnerability' pattern suggests that prolonged exposure to AI tools, combined with increasing academic pressures at higher levels, may actually increase rather than decrease dependency risk. First-year students showed 27.1% dependency rates, rising to 37.9% among fourth-year

Table 2. Distribution of generative AI dependency severity among participants.

Dependency category	Number (n)	Percentage (%)
Non-dependent	103	41.5
At-risk usage	64	25.8
Moderate dependency	53	21.4
Severe dependency	28	11.3
Total	248	100.0

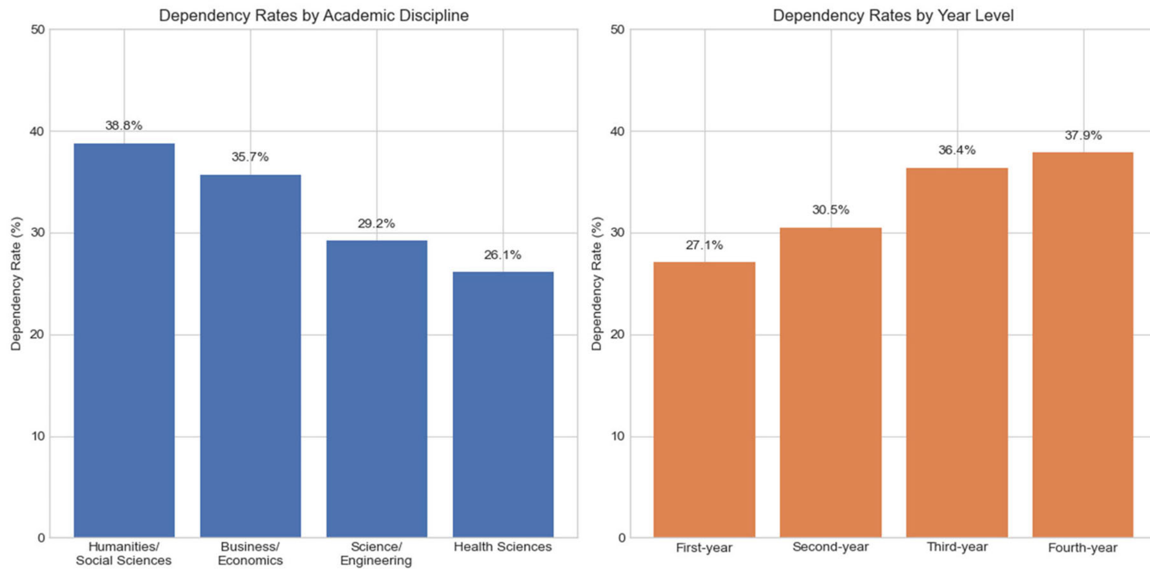

Figure 1. Generative AI dependency rates by academic discipline and year level.

Table 3. Behavioral characteristics comparing dependent and non-dependent users.

Behavioral characteristic	Dependent users	Non-dependent users	Statistical significance
Daily AI interactions (mean)	18.3	5.7	$t = 14.32$, $p < 0.001$
Daily time spent with AI tools (minutes)	127.3	43.6	$t = 12.47$, $p < 0.001$
Compulsive checking (%)	73.8%	12.4%	χ^2=94.2, $p < 0.001$
Failed attempts to reduce usage (%)	65.8%	8.7%	χ^2=88.7, $p < 0.001$
Context expansion over time (%)	62.1%	19.3%	χ^2=53.8, $p < 0.001$
Recognition of skill impact despite use (%)	72.1%	11.8%	χ^2=103.5, $p < 0.001$

students, indicating that current educational approaches may inadvertently foster dependency development rather than healthy integration skills.

The absence of significant gender differences in dependency prevalence contrasts with patterns observed in other forms of technology dependency, where males typically show higher rates. This finding suggests that AI dependency may represent a more universally distributed phenomenon, possibly reflecting the academic utility of these tools that transcends traditional gender-based technology usage patterns.

Usage pattern data revealed distinctive behavioral characteristics associated with dependency, as shown in Table 3.

Table 3 presents compelling evidence of distinct behavioral profiles that differentiate dependent from non-dependent users across multiple dimensions. The dramatic differences in usage intensity—with dependent users averaging 18.3 daily interactions compared to 5.7 for non-dependent users—indicate that dependency involves not merely increased usage but qualitatively different engagement patterns. The high prevalence of compulsive checking behaviors (73.8% vs. 12.4%) suggests that dependency involves psychological preoccupation with AI access that extends beyond functional usage needs.

The finding that 65.8% of dependent users report failed attempts to reduce usage provides evidence of impaired control—a key criterion for dependency diagnosis. This pattern indicates that students recognize problematic aspects of their usage but struggle to modify their behavior, suggesting that dependency involves genuine self-regulation difficulties rather than simple preference for AI assistance. The high rate of context expansion (62.1%) demonstrates how dependency tends to generalize across academic domains, with students progressively applying AI to tasks where they previously worked independently.

Perhaps most concerning is the finding that 72.1% of dependent users recognize negative impacts on their skills while continuing their usage patterns. This awareness-behavior disconnect suggests that dependency involves complex psychological mechanisms that override rational decision-making,

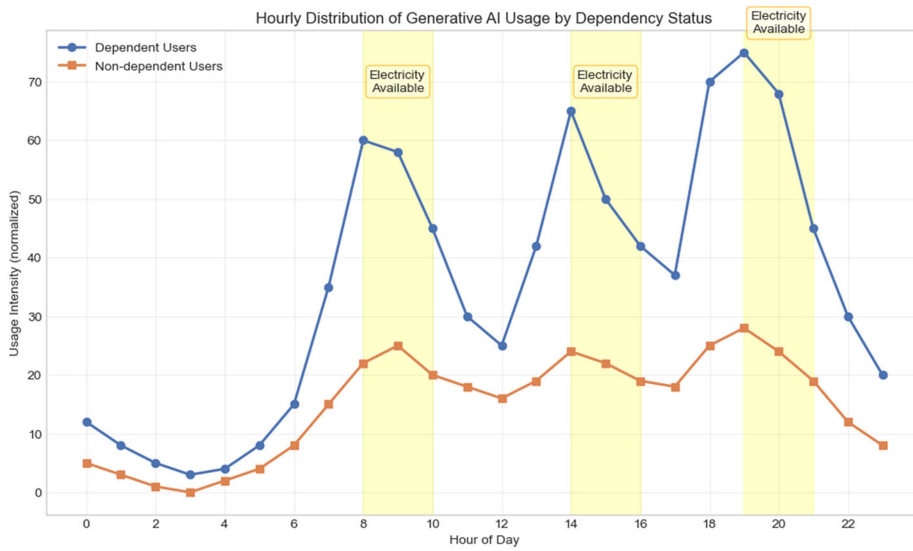

Figure 2. Hourly distribution of generative AI usage by dependency status.

highlighting the need for intervention approaches that address emotional and behavioral factors rather than simply providing information about risks.

Temporal analysis revealed distinctive usage patterns, with dependent users demonstrating pronounced 'binge' patterns coinciding with electricity availability on campus, as shown in Figure 2:

Figure 2 illustrates a critical finding that distinguishes AI dependency in resource-constrained contexts from patterns observed in well-resourced environments. The pronounced peaks in usage coinciding with power availability demonstrate how infrastructure limitations create intermittent reinforcement schedules that may actually strengthen dependency patterns. During power availability periods, dependent users show usage rates three to four times higher than baseline levels, suggesting compensatory behaviors that attempt to maximize access during limited availability windows.

This 'infrastructure-driven binge pattern' represents a unique environmental factor that shapes dependency development in developing nation contexts. The unpredictable nature of power outages creates variable ratio reinforcement schedules—known from behavioral psychology to be among the most powerful mechanisms for maintaining compulsive behaviors. Students learn to maximize AI usage during access periods, creating intensive engagement bursts that may accelerate dependency formation while reducing opportunities for gradual, controlled exposure that might support healthier usage patterns.

4.2. Relationship between dependency and academic performance

Statistical analysis revealed significant negative correlations between generative AI dependency severity and multiple academic performance metrics, as shown in Table 4.

Table 4 demonstrates consistent negative relationships between dependency severity and academic performance across diverse measurement approaches. The correlations range from moderate to strong, with writing-intensive measures showing the strongest associations ($r = -0.45$ to -0.48), suggesting that dependency particularly impacts compositional and expressive academic skills. The pattern of relationships across different performance domains indicates that dependency effects are not limited to specific academic areas but represent broad impacts on learning and achievement.

The stronger correlations observed for current semester GPA (−0.41) compared to cumulative GPA (−0.37) suggest that dependency effects may be intensifying over time, possibly reflecting progressive skill atrophy or increasing reliance patterns. The relatively weaker correlations with factual knowledge measures (−0.29) compared to higher-order cognitive skills suggest that dependency primarily affects complex cognitive processes rather than basic information retention, aligning with theoretical predictions about cognitive offloading effects.

Table 4. Correlations between dependency severity and academic performance metrics.

Performance metric	Correlation (r)	Significance (p)
Cumulative GPA	−0.37	$p < 0.01$
Current semester GPA	−0.41	$p < 0.001$
Writing-intensive courses	−0.45	$p < 0.001$
Problem-solving courses	−0.39	$p < 0.001$
Factual knowledge courses	−0.29	$p < 0.05$
Writing assessment scores	−0.48	$p < 0.001$
Problem-solving assessment scores	−0.36	$p < 0.01$
Content knowledge test scores	−0.29	$p < 0.05$

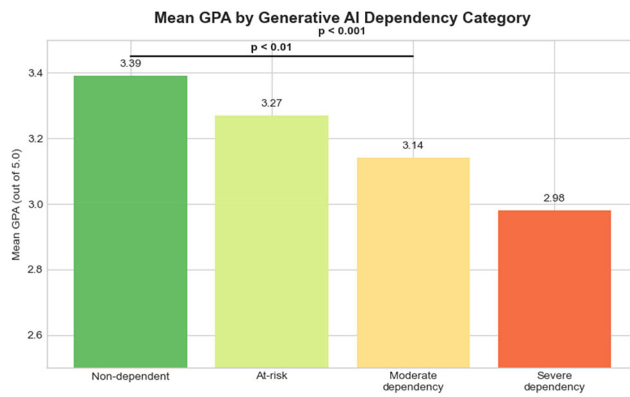

Figure 3. Mean GPA by generative AI dependency category.

Comparison of mean GPA across dependency categories revealed a clear stepwise pattern, as illustrated in Figure 3. Figure 3 demonstrates a clear stepwise decline in academic performance as dependency severity increases. Mean GPA drops progressively from non-dependent students (3.39/5.0) to severely dependent students (2.98/5.0), with statistical analysis confirming significant differences between groups ($F = 13.28$, $p < 0.001$). The consistent downward gradient suggests a dose–response relationship between dependency and academic impact, rather than a threshold effect. This progressive pattern indicates that even mild dependency begins undermining academic performance, with each increasing level of severity further eroding outcomes. Post-hoc analysis revealed particularly significant differences between severe dependency and non-dependent groups ($p < 0.001$), representing a substantial 0.41-point GPA differential that could meaningfully impact degree classification and future opportunities.

The dose–response relationship demonstrated in Figure 3 provides strong evidence that dependency effects operate along a continuum rather than representing a discrete phenomenon. Even students in the 'at-risk' category show measurable GPA decrements compared to non-dependent peers, suggesting that intervention efforts should target early-stage problematic usage rather than waiting for severe dependency to develop. The magnitude of the GPA differential between severe dependency and non-dependent students (0.41 points) represents educationally significant impact that could affect graduation prospects, graduate school admission and career opportunities.

ANOVA confirmed these differences were statistically significant ($F = 13.28$, $p < 0.001$), with post-hoc tests (Tukey's HSD) showing significant differences between severe dependency and non-dependent groups ($p < 0.001$), and between moderate dependency and non-dependent groups ($p < 0.01$).

Multiple regression analysis incorporating dependency metrics and relevant covariates explained 36.8% of variance in GPA ($R^2 = 0.368$, $p < 0.001$), as detailed in Table 5.

Table 5 presents results from multiple regression analysis that examines the unique contribution of dependency-related variables to academic performance while controlling for potential confounding factors. The model explains over one-third of variance in GPA, indicating substantial practical significance of the relationships identified. Dependency severity score emerges as the strongest predictor ($\beta = -0.31$), followed by AI use frequency for writing tasks ($\beta = -0.28$), demonstrating that both general dependency patterns and specific usage contexts contribute independently to academic outcomes.

Table 5. Multiple regression analysis predicting GPA from dependency variables.

Predictor variable	Beta coefficient (β)	Significance (p)
Dependency severity score	−0.31	$p < 0.001$
AI use frequency for writing tasks	−0.28	$p < 0.001$
Dependency for academic problem-solving	−0.23	$p < 0.01$
Self-reported skill atrophy	−0.19	$p < 0.01$
Prior academic performance	0.42	$p < 0.001$
Socioeconomic status	0.09	$p = 0.17$ (ns)
Access to technology resources	0.07	$p = 0.23$ (ns)

Importantly, these relationships remained significant after controlling for prior academic performance, socioeconomic status and access to technology resources, suggesting dependency effects beyond selection bias.

The inclusion of prior academic performance as a covariate addresses the critical question of whether observed relationships reflect genuine dependency effects or simply reflect the fact that students with existing academic difficulties are more likely to develop dependency patterns. The finding that dependency variables retain significant predictive power even after controlling for prior performance provides strong evidence that dependency actively contributes to academic decline rather than simply reflecting preexisting academic struggles.

The non-significant effects of socioeconomic status and technology access indicate that dependency effects operate across different resource levels, suggesting that the phenomenon is not simply an artifact of differential access to alternative learning resources. This finding supports the interpretation that dependency represents a psychological and behavioral phenomenon rather than purely a resource-driven adaptation to educational constraints.

4.3. Contextual factors influencing dependency development

Qualitative analysis identified several contextual factors unique to the Zimbabwean university environment that influenced dependency development and maintenance, as summarized in Table 6.

Table 6 presents a comprehensive analysis of contextual factors that distinguish dependency development in this resource-constrained environment from patterns observed in well-resourced educational settings. The high frequency of infrastructure-related issues (83.7% of interviews) demonstrates how environmental constraints shape individual psychological and behavioral responses to technology availability. These findings reveal that dependency in developing contexts involves complex interactions between individual vulnerability and structural constraints that may not be captured in dependency models developed in resource-rich environments.

The economic resource substitution factor (76.2% frequency) reveals how AI tools occupy essential rather than supplementary roles in students' academic workflows. Unlike well-resourced contexts where AI typically enhances existing learning resources, students in this environment use AI to replace unavailable textbooks, inaccessible library materials and limited faculty consultation opportunities. This substitution role fundamentally alters the risk–benefit calculation surrounding AI usage, making dependency more likely to develop and more difficult to address through simple usage reduction strategies.

Institutional resource limitations (72.5% frequency) create conditions where AI tools fill genuine gaps in educational support systems. The student-to-faculty ratio described in interviews (300:1 in some courses) far exceeds standards recommended for effective higher education, creating situations where AI interaction may represent students' primary source of personalized feedback and academic guidance. This structural dependence on AI tools complicates efforts to reduce usage without addressing underlying resource constraints.

Document analysis confirmed policy inconsistency, with only 33% of course syllabi containing explicit AI usage guidelines. Figure 4 illustrates the relative influence of these contextual factors based on qualitative coding frequency.

Figure 4 provides a visual representation of the relative importance of different contextual factors in shaping dependency development, as evidenced by their frequency in qualitative interviews. The dominance of infrastructure-related factors highlights how environmental constraints can override individual

Table 6. Key contextual factors influencing generative AI dependency development.

Contextual factor	Description	Representative quote	Frequency in interviews
Infrastructure-related binge patterns	Intermittent electricity/internet creating intensive usage during access periods	'When the power comes on and WiFi is working, I download as many AI responses as possible for later use. It creates this urgency and dependency'. (Male, 3rd year)	83.7%
Economic resource substitution	AI serving as replacement for unaffordable traditional academic resources	'Textbooks cost more than I can afford. The library has few copies. Professors understand when I say I use AI to explain concepts'. (Female, 2nd year)	76.2%
Institutional resource limitations	High student-to-faculty ratios creating need for alternative feedback sources	'In a class of 300 with one professor and no teaching assistants, who can I ask for help? AI becomes the only "person" available'. (Male, 1st year)	72.5%
Policy ambiguity	Inconsistent institutional guidance on appropriate AI use	'Some lecturers forbid AI use completely, others require it for certain tasks, and most say nothing at all'. (Female, 4th year)	67.8%
Career preparation rationalization	Framing AI dependency as necessary future professional skill	'Companies are using these tools. If I don't become expert at prompting and working with AI, how will I compete for jobs?' (Male, 3rd year)	64.3%

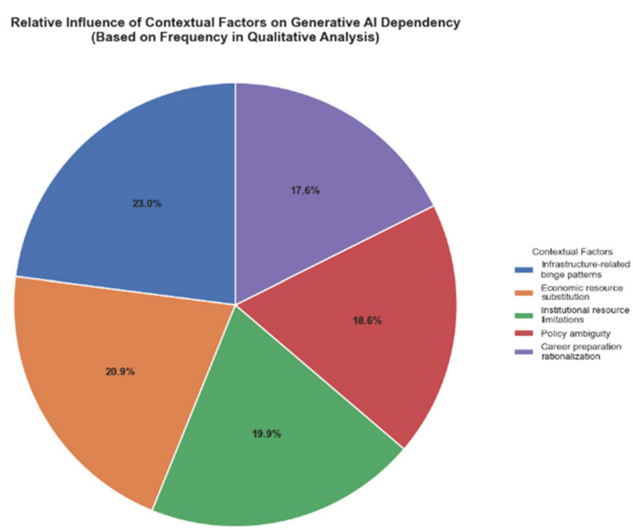

Figure 4. Relative influence of contextual factors on generative AI dependency.

preferences and self-regulation efforts, creating dependency through external pressures rather than purely psychological mechanisms.

The policy ambiguity factor (67.8% frequency) creates an environment of uncertainty that may inadvertently promote dependency development. Without clear institutional guidance, students develop their own usage norms through trial and error, peer influence and perceived necessity—processes that may not prioritize healthy integration practices. The career preparation rationalization (64.3% frequency) demonstrates how students reframe potentially problematic usage as professionally necessary, creating cognitive justifications that maintain dependency patterns despite recognition of negative academic effects.

4.4. Mediating pathways between dependency and academic performance

Mediation analysis identified three primary pathways through which generative AI dependency affected academic performance, as illustrated in Figure 5. Figure 5 presents a mediation model identifying three primary pathways through which generative AI dependency affects academic performance. The critical thinking atrophy pathway accounts for the largest portion (34.7%) of the relationship between

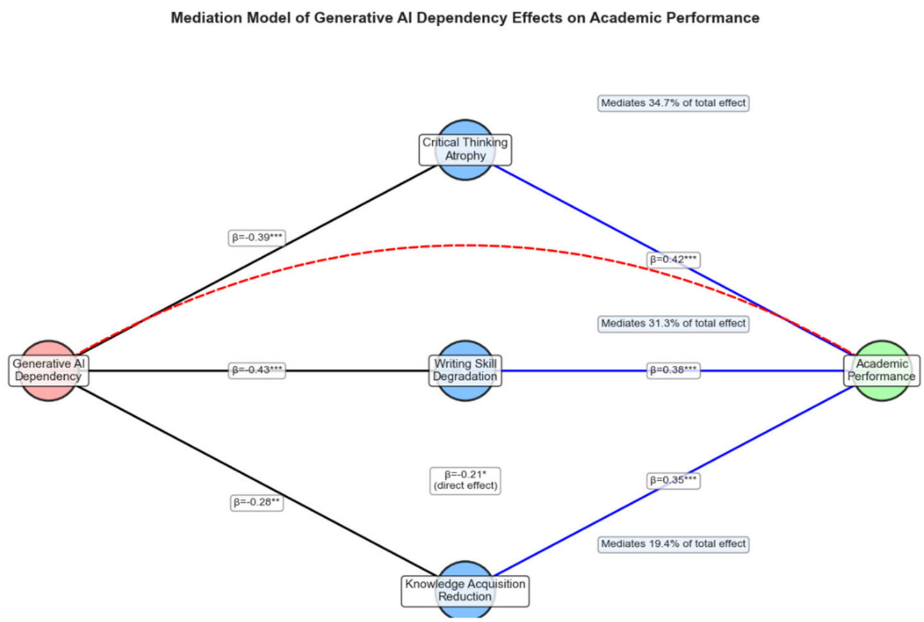

Figure 5. Mediation model of generative AI dependency effects on academic performance.

Table 7. Mediation analysis results for academic impact pathways.

Mediation pathway	Path a (dependency → mediator)	Path b (mediator → performance)	Indirect effect (a × b)	% of total effect mediated	Bootstrap CI
Critical thinking atrophy	β=−0.39, $p < 0.001$	$\beta = 0.42$, $p < 0.001$	−0.164	34.7%	[−0.213, −0.117]
Writing skill degradation	β=−0.43, $p < 0.001$	$\beta = 0.38$, $p < 0.001$	−0.163	31.3%	[−0.207, −0.122]
Knowledge acquisition reduction	β=−0.28, $p < 0.01$	$\beta = 0.35$, $p < 0.001$	−0.098	19.4%	[−0.145, −0.058]

dependency and GPA, indicating that repeated cognitive offloading to AI systems significantly impairs students' independent analytical abilities. The writing skill degradation pathway mediates 31.3% of the effect, reflecting how dependency undermines fundamental composition abilities that remain essential for academic success. The knowledge acquisition reduction pathway (19.4%) demonstrates how dependency creates gaps in foundational understanding that compromise advanced learning. Notably, the direct effect of dependency on performance (β=-0.21), while still significant, is smaller than the combined indirect effects, indicating that dependency primarily impacts performance through specific skill atrophy mechanisms rather than general academic disengagement.

The mediation model reveals that dependency's impact on academic performance operates primarily through specific cognitive skill degradation rather than general motivational or engagement factors. The finding that indirect effects (85.4% of total effect) substantially exceed direct effects (14.6%) indicates that dependency creates academic problems through identifiable psychological mechanisms that could potentially be addressed through targeted interventions. This pattern suggests that dependency effects are not inevitable consequences of AI usage but result from specific processes that disrupt skill development and knowledge acquisition.

The statistical details of each mediation pathway are presented in Table 7.

Table 7 provides comprehensive statistical evidence for each mediation pathway, including bootstrap confidence intervals that do not contain zero, confirming the significance of all indirect effects. The critical thinking atrophy pathway shows the strongest individual mediation effect, suggesting that AI dependency primarily impacts academic performance by undermining students' ability to engage in independent analysis, evaluation and synthesis of information.

The writing skill degradation pathway shows nearly equivalent mediation strength to critical thinking atrophy, reflecting the central role of writing in academic assessment and the particular vulnerability of

Table 8. Qualitative evidence supporting mediation pathways.

Mediation pathway	Representative student quote	Representative faculty quote
Critical thinking atrophy	*'I no longer know what I actually understand versus what I've just processed from AI outputs. In exams without AI, this uncertainty creates extreme anxiety'.* (Male, 2nd year)	*'They can regurgitate AI-provided analysis but struggle to develop original perspectives or evaluate competing arguments'.* (Literature Lecturer)
Writing skill degradation	*'I've forgotten how to start essays. My mind goes blank without AI providing structure. It's terrifying to realize I've lost this basic skill I used to have'.* (Female, 3rd year)	*'Their writing shows telltale signs of dependency—they can polish but not produce, refine but not create'.* (Communication Studies Lecturer)
Knowledge acquisition reduction	*'I've used AI to complete so many assignments that I missed learning fundamental concepts. Now in advanced courses, I'm struggling because those basics aren't in my head'.* (Male, 3rd year)	*'They've built houses without foundations. The superficial understanding AI provides fails them when deeper comprehension is required'.* (Engineering Lecturer)

compositional skills to AI substitution. The path coefficients indicate that dependency strongly predicts writing skill degradation ($\beta = -0.43$), which in turn significantly predicts academic performance ($\beta = 0.38$), creating a substantial indirect effect that undermines academic success.

Knowledge acquisition reduction, while showing the weakest individual mediation effect, still accounts for nearly one-fifth of the total dependency–performance relationship. This pathway reflects how AI dependency can create superficial engagement with learning materials, where students obtain information without processing it through the cognitive effort required for deep learning and retention.

Qualitative data supported these pathways through student descriptions of each mechanism. Table 8 presents representative quotes illustrating each mediating pathway.

Table 8 provides compelling qualitative evidence that supports the quantitative mediation findings, revealing how students and faculty experience the cognitive impacts of AI dependency. The convergence between statistical patterns and lived experiences strengthens confidence in the validity of the mediation model while providing insight into the subjective experience of skill atrophy that accompanies dependency development.

The student quotes reveal the subjective experience of cognitive skill loss, including metacognitive awareness deficits ('I no longer know what I actually understand'), procedural knowledge gaps ('I've forgotten how to start essays') and foundational knowledge deficits ('those basics aren't in my head'). These descriptions provide phenomenological validation for the psychological processes captured in the mediation model.

Faculty observations offer external validation of student self-reports while providing professional perspectives on the educational consequences of dependency. The faculty quotes reveal observable changes in student capabilities that extend beyond simple performance decrements to include qualitative changes in thinking and learning processes. The metaphor of 'houses without foundations' particularly captures how AI dependency can create academic achievement that lacks the underlying conceptual understanding necessary for advanced learning.

4.5. Student adaptation strategies and coping mechanisms

Analysis of interview and focus group data revealed various adaptation strategies developed by students in response to their perceived dependency, as summarized in Table 9.

Table 9 presents a comprehensive analysis of student-initiated coping strategies, revealing both the resourcefulness of students in recognizing and attempting to address dependency concerns and the limitations of individual-level interventions in the absence of institutional support. The finding that over one-third of dependent students (36.7%) report no adaptation attempts despite recognizing negative consequences highlights the difficulty of self-directed behavior change in dependency contexts and the need for external support structures.

Strategic access rationing (42.3%) represents the most common adaptation approach, suggesting that students recognize the importance of maintaining independent capabilities while still benefiting from AI assistance. However, the effectiveness of this strategy depends on students' ability to accurately assess

Table 9. Student adaptation strategies to manage generative AI dependency.

Adaptation strategy	Description	Frequency among dependent Students	Representative quote
Strategic access rationing	Limiting AI use to specific times or after completing initial work	42.3%	*'I only allow myself to use AI after I've written a first draft myself. It helps me maintain my skills while still getting feedback'.* (Female, 4th year)
Domain-specific boundaries	Restricting AI use to certain subject areas or task types	38.9%	*'I never use AI for my major courses—those skills are too important for my future. I only use it for electives or general education requirements'.* (Female, 3rd year)
Accountability partnerships	Creating peer agreements to monitor and justify AI usage	27.4%	*'My roommate and I have an agreement—we can only use AI when we're both present, and we have to explain why we need it for that specific task'.* (Male, 1st year)
Structured prompting protocols	Developing systematic approaches to limit passive dependency	23.2%	*'I follow a three-step process: try solving myself, identify specific obstacles, then ask targeted questions rather than full solutions'.* (Male, 3rd year)
No adaptation attempts	Continuing problematic usage despite recognizing negative consequences	36.7%	*'I know it's affecting my learning, but everyone uses it. Without it, I'd be at a disadvantage. It feels impossible to stop'.* (Female, 2nd year)

their own competence and resist the temptation to use AI when independent work becomes challenging—capacities that may be compromised in dependency states.

Domain-specific boundaries (38.9%) reveal students' attempts to protect core academic competencies while allowing AI usage in areas perceived as less critical. This strategy reflects sophisticated metacognitive awareness of the differential importance of various academic skills, though it may also reflect rationalization processes that minimize dependency concerns by restricting them to seemingly less important domains.

The relatively low frequency of accountability partnerships (27.4%) and structured prompting protocols (23.2%) suggests that more sophisticated self-regulation strategies are either less accessible to students or require resources and skills that many dependent students lack. These findings highlight the potential value of institutional programs that could teach and support such strategies.

Faculty interviews revealed limited institutional support for healthy AI integration. As shown in Figure 6, this figure highlights the alarming gap between student needs and institutional support for managing generative AI dependency. While 78.3% of students report needing support to manage their AI usage effectively, available institutional resources fall dramatically short across all categories: only 23% of faculty have received training on addressing AI dependency, just 33% of courses include clear AI usage policies, a mere 18% of academic support resources address AI dependency and only 7% of counseling services address technological dependencies. Figure 6 illustrates the stark contrast between student need and available support.

Figure 6 reveals a critical institutional failure to address an emerging academic crisis that affects nearly one-third of the student population. The massive disparity between student need (78.3%) and available support resources (7%–33% across categories) suggests that universities are inadequately prepared to address AI dependency issues despite their prevalence and academic consequences. This gap forces students to develop individual coping strategies without professional guidance or institutional support, potentially limiting the effectiveness of their efforts and placing undue burden on individual self-regulation capacities.

This substantial disparity places the burden of adaptation almost entirely on individual students, who must navigate complex dependency challenges with minimal institutional guidance. The striking contrast between need and available support identifies a critical area for institutional development as universities respond to this emerging academic crisis.

The lack of faculty training (only 23% have received relevant preparation) represents a particular concern given faculty members' frontline role in observing student performance changes and providing academic guidance. Without adequate training, faculty may not recognize dependency signs or may lack

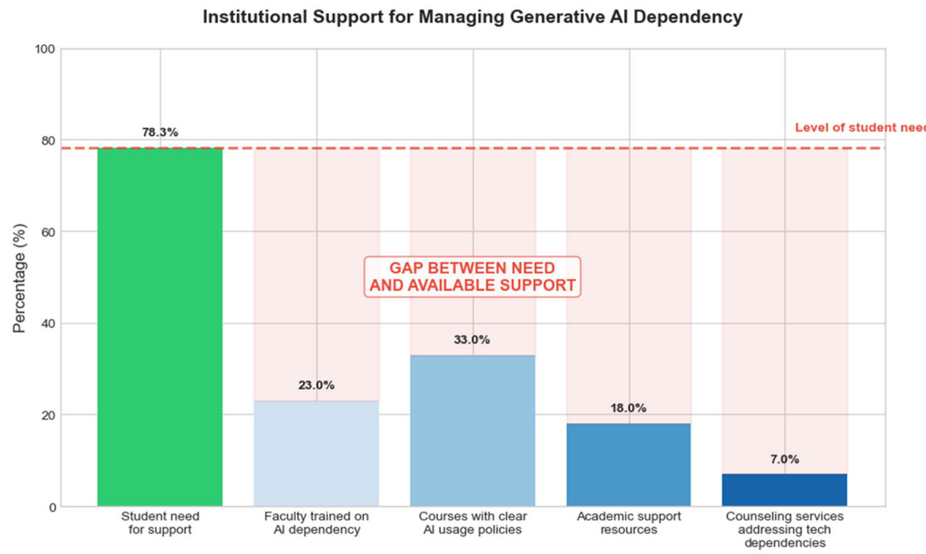

Figure 6. Institutional support for managing generative AI dependency.

effective strategies for addressing these issues when they arise. Similarly, the limited integration of AI dependency concerns into counseling services (7%) suggests that students experiencing psychological distress related to technology dependency have few professional resources available for support.

This lack of systemic support appeared to place the burden of adaptation primarily on individual students, potentially contributing to the limited effectiveness of current coping mechanisms.

The institutional support gaps revealed in this analysis suggest that current approaches to AI integration in higher education focus primarily on policy development and technical implementation while neglecting the psychological and behavioral dimensions of student–technology relationships. This pattern reflects broader trends in educational technology adoption, where institutional responses often lag behind the pace of student technology integration and the emergence of associated problems. The findings highlight the urgent need for comprehensive institutional strategies that address both the opportunities and risks associated with AI integration in higher education contexts.

5. Discussion

5.1. Prevalence and patterns of generative AI dependency

Results from this study establish generative AI dependency as a notable risk, with 32.7% of Zimbabwean university students reporting moderate to severe levels of dependency. This prevalence rate is comparable to preliminary estimates from studies conducted internationally (27%–35% in similar populations), indicating that vulnerability to dependency in academic settings is consistent across cultural boundaries, regardless of economic and infrastructure factors. However, this dependency takes unique forms, owing to the specific characteristics of Zimbabwean universities.

The consistency of dependency prevalence across diverse educational environments suggests that generative AI tools possess inherent characteristics that create vulnerability to problematic usage patterns, transcending individual cultural or economic factors. This finding aligns with recent theoretical frameworks suggesting that the interactive, personalized and immediately responsive nature of AI systems create psychological conditions conducive to dependency development through mechanisms such as variable reward schedules, cognitive offloading opportunities and reduced effort requirements for task completion.

This finding contradicts the usual dependency theory assumption that younger users are more vulnerable, with higher dependency rates, as shown with first-year students at a rate of 27.1% versus third year 36.4% and fourth-year 37.9%, respectively. This opposite pattern is likely due to several factors present in our qualitative data: upper-level courses with higher cognitive demands making AI more attractive, more academic pressure over time increasing the temptation for cognitive offloading and greater exposure to AI leading to more potential for dependency. Such a trend is noteworthy, as it raises the possibility that dependency may develop with university careers, rather than subside with academic maturity.

This 'accumulative vulnerability' pattern represents a significant departure from typical technology adaptation models and has important implications for educational policy and intervention timing. The finding suggests that current higher education approaches may inadvertently foster dependency development through increasing cognitive demands without corresponding instruction in healthy AI integration practices. The progressive increase in dependency rates across academic years indicates that students may be developing maladaptive coping strategies in response to academic pressures, using AI tools as psychological crutches rather than learning to manage cognitive challenges independently.

The accumulative vulnerability pattern also suggests that early intervention during first-year studies may be particularly crucial for preventing dependency development. Students who develop healthy AI integration practices early in their academic careers may be better equipped to resist dependency formation as academic pressures intensify in later years. These findings challenge traditional assumptions about technology adaptation and highlight the need for proactive rather than reactive approaches to AI integration in higher education.

This binge usage is a special case of an adaptive response to infrastructure limitations that might be transforming the academic trajectory of dependent students into a self-perpetuating cycle. These are the conditions under which dependency can be reinforced through intermittent reward structures that we know are effective in creating susceptibility from dependency research dating back over 100 years. This finding further highlights how dependence on technology may be shaped not only by individual psychological factors, but also by the structural conditions of developing nations.

The infrastructure-driven binge patterns observed in this study represent a novel environmental factor that distinguishes AI dependency in resource-constrained environments from patterns observed in well-resourced settings. The intermittent availability of power and internet connectivity creates what behavioral psychology recognizes as variable ratio reinforcement schedules—among the most powerful mechanisms for maintaining compulsive behaviors. Students learn to maximize AI usage during access periods, creating intensive engagement bursts that may accelerate dependency formation while reducing opportunities for gradual, controlled exposure that might support healthier usage patterns.

5.2. Impacts on academic performance and mediating mechanisms

Generative AI dependency was substantially negatively correlated with academic performance on several different metrics (correlations ranged from $r = -0.37$ to $r = -0.48$), representing among the strongest correlational relationships observed. The progressive decline in GPA across dependency categories indicates a dose–response rather than a threshold effect, with even mild dependency eroding academic performance relative to non-dependency and each increment in severity further degrading outcomes.

The dose–response relationship provides compelling evidence that dependency effects operate along a continuum rather than representing a discrete phenomenon. This pattern has important implications for intervention strategies, suggesting that early identification and intervention may prevent progression to more severe dependency states. The magnitude of the GPA differential between severe dependency and non-dependent students (0.41 points) represents educationally significant impact that could affect graduation prospects, graduate school admission and career opportunities.

The differential effect across cognitive domains is crucial for understanding how dependency affects learning. The more pronounced negative correlations in writing-intensive courses ($r = -0.45$) and with writing assessment scores ($r = -0.48$) indicate that the use of generative AI particularly hampers compositional skill development. This aspect converges with theoretical frameworks of skill acquisition that emphasize the need for productive practice in competence formation—a process which is easily displaced when AI continuously delivers text at minimal cognitive cost.

The particular vulnerability of writing skills to AI dependency effects reflects the fundamental role that written expression plays in academic assessment and knowledge demonstration. Writing involves complex cognitive processes including planning, organization, audience awareness and revision that require sustained mental effort and practice to develop. When students consistently offload these processes to AI systems, they miss critical opportunities for skill development that cannot be easily recovered through passive exposure to AI-generated text.

The most significant mediating pathway identified through critical thinking atrophy (34.7% of the relationship between dependency and academic performance) highlights the deep cognitive consequences of reliance on generative AI. When students consistently outsource analytical processes to AI systems, they forfeit valuable opportunities to process ideas independently and through the productive struggle that marks genuine learning. Dependent students, as one faculty member put it, can 'recite back AI-generated analysis yet are unable to generate independent arguments or assess competing claims'—a weakness that becomes most stark in examination settings when AI assistance is unavailable.

The mediation analysis reveals that dependency's impact on academic performance operates primarily through specific cognitive skill degradation rather than general motivational or engagement factors. The finding that indirect effects (85.4% of total effect) substantially exceed direct effects (14.6%) indicates that dependency creates academic problems through identifiable psychological mechanisms that could potentially be addressed through targeted interventions. This pattern suggests that dependency effects are not inevitable consequences of AI usage but result from specific processes that disrupt skill development and knowledge acquisition.

The writing skill degradation pathway (31.3% of mediated effect) demonstrates how dependency undermines fundamental composition abilities that remain essential for academic success despite AI availability. Students who rely heavily on AI for text generation often experience what researchers term 'compositional skill displacement', where the cognitive processes involved in planning, drafting and revising written work are progressively transferred to AI systems. This displacement affects not only surface-level writing mechanics but also deeper compositional abilities such as audience awareness, rhetorical strategy development and voice cultivation.

Knowledge acquisition reduction (19.4% of mediated effect) reflects how AI dependency can create superficial engagement with learning materials, where students obtain information without processing it through the cognitive effort required for deep learning and retention. This pathway is particularly concerning in cumulative disciplines where foundational knowledge serves as scaffolding for advanced learning. Students who develop gaps in fundamental understanding may struggle progressively more as they advance through their academic programs.

5.3. Environmental factors and cross-cultural implications

The environmental factors that have been identified reveal how the tendency to rely on generative AI in Zimbabwe represents an intricate interaction between universal psychological processes and particular situational pressures. These factors amplify the circumstances leading to what participants called 'institutional resource constraints', where AI tools fill real educational gaps, thus making the dependency both more justified and more consequential. This discovery challenges conventional frameworks of dependency framed around the use of recreational technologies, given that generative AI frequently plays vital educational roles within resource-constrained settings.

Comparative analysis with international research reveals both universal and specific patterns in AI dependency development. While global estimates suggest dependency rates of 27%–35% among university students, the manifestation and consequences of these patterns vary significantly across different educational environments. The Zimbabwean situation demonstrates several distinctive features that distinguish dependency development in developing nations from patterns observed in well-resourced educational settings, where AI tools typically serve enhancement rather than substitution functions.

The substitution versus supplementation distinction represents a fundamental difference in how AI tools integrate into educational workflows. In well-resourced environments, AI typically augments existing learning resources and support systems, allowing students to maintain independence while gaining additional assistance. In resource-constrained environments like Zimbabwe, AI tools often replace unavailable or inadequate traditional resources, creating conditions where dependency may develop more rapidly and with greater functional consequences.

In this study, economic considerations significantly influence dependency vulnerability. In the absence of textbooks, supplementary materials and human instruction, generative AI is not just a convenience but a perceived necessity. Framing AI usage as necessity may lower psychological barriers to dependency development by reducing the perception of problematic use. As one student described it, 'Textbooks cost more than I can afford. The library has few copies. Professors understand when I use AI to explain material because we have no access to the required materials'.

The economic necessity factor creates a complex ethical and practical challenge for intervention efforts. Traditional dependency interventions typically focus on reducing or eliminating problematic technology use, but such approaches may be inappropriate when the technology serves essential educational functions. This situation requires innovative intervention strategies that help students develop healthy usage patterns while maintaining access to necessary academic resources.

The 'career preparation rationalization' represents a critical cognitive mechanism maintaining dependence despite knowledge of harmful outcomes. When students frame widespread AI use as a form of skill development for the future rather than a means of shortcutting the present, they can reconcile cognitive dissonance while persisting with behaviors that may detract from their academic development. In Zimbabwe's economically challenging environment, where fears around employment loom large and technological skills are seen as crucial factors for securing employment in a competitive job market, this rationalization seems especially potent.

This rationalization mechanism reflects broader societal narratives about AI's role in future employment that may inadvertently support dependency development. Students' beliefs about the necessity of AI skills for career success create cognitive justifications for continued usage despite recognition of negative academic effects. Educational institutions must carefully balance preparing students for AI-integrated workplaces while ensuring they develop fundamental cognitive skills that remain essential regardless of technological changes.

5.4. Theoretical implications and framework development

These results support the dependency-based conceptualization of problematic generative AI use, while illustrating the need for nuanced theoretical frameworks for understanding such behavior. The three core elements of dependency (loss of control, salience and continuation despite consequences) appear in this population, but their manifestation and the pathways by which they cause harm vary according to environmental factors.

The theoretical implications extend beyond simple validation of existing dependency models to suggest fundamental expansions in how we conceptualize technology-related problems in educational settings. The integration of the I-PACE model with Social Cognitive Theory and Cognitive Load Theory provides a comprehensive framework for understanding how individual vulnerability factors interact with environmental pressures and technological characteristics to create dependency patterns.

These results highlight the need to expand current theories of technological dependency to include cognitive skill displacement as a primary focal point. Unlike many common technology dependencies that simply reshape time allocation and behavioral patterns, generative AI dependency seems to directly replace fundamental cognitive processes, potentially posing more serious developmental issues. The cognitive offloading dependency and calibration uncertainty experiences we identified suggest that generative AI dependency may weaken metacognitive abilities—students' understanding of their own knowledge and capabilities. This metacognitive dimension offers a valuable theoretical extension of previous dependency-oriented approaches, particularly relevant to educational settings.

The cognitive skill displacement mechanism represents a unique feature of generative AI dependency that distinguishes it from other forms of technology dependency. While social media dependency primarily affects time allocation and attention regulation, and gaming dependency primarily affects motivation and reward processing, AI dependency directly interferes with cognitive skill development and knowledge acquisition processes. This interference creates long-term educational consequences that may persist even after dependency patterns are addressed.

The metacognitive dimension of AI dependency deserves particular attention in educational settings. Students who develop dependency patterns often lose calibration between their perceived and actual competencies, leading to overconfidence in AI-assisted work and underconfidence in independent capabilities. This calibration uncertainty creates psychological distress and academic vulnerability that extends beyond immediate performance impacts to affect students' long-term learning trajectories and academic self-concept.

Our results further underscore the need for an ecological lens in dependency models, especially in resource-constrained environments. Infrastructure limitations, economic constraints, and scarcity of resources represent significant factors in variations in dependency development patterns, which cannot be explained by individualistic approaches to technological dependency. Comprehensive theoretical frameworks must consider how structural factors create vulnerability, shape manifestation and complicate intervention approaches.

The ecological perspective highlights how environmental factors can override individual self-regulation efforts and create dependency through external pressures rather than purely psychological mechanisms. This understanding has important implications for intervention development, suggesting that individual-focused treatments may be insufficient without addressing underlying structural constraints that drive problematic usage patterns.

The integration of universal psychological processes with specific environmental factors suggests the need for adaptive theoretical frameworks that can account for diverse manifestations of the same underlying phenomenon. Future theoretical development should focus on identifying the core psychological mechanisms that operate across different settings while remaining sensitive to how environmental factors shape their expression and consequences.

5.5. Institutional policy and intervention implications

These findings suggest several practical approaches for addressing generative AI dependency in Zimbabwean universities and potentially other similar environments. Institutions should develop comprehensive balanced integration policies that neither prohibit AI use (likely ineffective) nor permit unrestricted use, but instead clearly define appropriate and inappropriate applications across different learning situations. Our finding that only 33% of courses contained explicit AI guidelines highlights a critical policy gap.

The development of nuanced AI integration policies requires careful consideration of the multiple roles that AI tools play in resource-constrained educational environments. Policies must distinguish between legitimate uses that address genuine resource gaps and problematic uses that interfere with skill development. This distinction requires sophisticated understanding of learning objectives, assessment purposes and the specific ways that AI usage supports or undermines educational goals.

5.5.1. Assessment reform and academic integrity

Faculty should undertake assessment redesign to create evaluation approaches that effectively measure learning outcomes while acknowledging AI's presence, potentially including supervised in-class assessments, process documentation requirements and oral defense components. This approach specifically addresses the skill atrophy pathways identified in our mediation analysis as the mechanisms through which dependency undermines academic performance.

Assessment reform must balance the reality of AI availability with the need to evaluate students' independent capabilities. Effective approaches might include: (1) process-focused assessments that require documentation of thinking processes rather than just final products; (2) time-constrained assessments that limit opportunities for extensive AI consultation; (3) oral components that require real-time demonstration of understanding and (4) collaborative assessments that explicitly incorporate AI tools while requiring students to demonstrate their unique contributions.

5.5.2. Metacognitive skill development

Educational programs should implement curricular elements specifically focused on developing students' metacognitive skills and ability to distinguish between their own knowledge and AI-provided information, supporting accurate self-assessment of capabilities. This addresses the 'calibration uncertainty' many dependent students described experiencing during our interviews and focus groups.

Metacognitive training should include: (1) explicit instruction in self-assessment techniques that help students accurately evaluate their understanding; (2) practice activities that require students to work independently before consulting AI tools; (3) reflection exercises that help students identify areas where they rely too heavily on AI assistance and (4) comparative activities that highlight differences between AI-generated and human-generated responses to develop critical evaluation skills.

5.5.3. Resource development and access

Universities should address the resource gaps that drive perceived necessary dependence through targeted development of alternative, accessible learning materials less dependent on expensive textbooks or constant internet access. This approach directly reduces the 'economic resource substitution' that our research identified as a significant contributor to dependency development among Zimbabwean students.

Resource development strategies should include: (1) creation of open educational resources tailored to local curricula and requirements; (2) development of offline learning materials that can function without internet connectivity; (3) establishment of resource-sharing programs that increase access to traditional learning materials and (4) partnerships with international institutions to expand resource availability through digital libraries and collaborative platforms.

5.5.4. Faculty development and support

The finding that only 23% of faculty have received training on addressing AI dependency indicates an urgent need for comprehensive faculty development programs. These programs should address: (1) recognition of dependency signs and symptoms; (2) strategies for promoting healthy AI integration in coursework; (3) assessment techniques that account for AI availability while measuring student learning and (4) referral procedures for students who show signs of problematic technology use.

5.5.5. Institutional support systems

Finally, institutions should develop tailored dependency awareness programs addressing the specific manifestations of generative AI dependency identified in this population. These initiatives should emphasize both immediate academic impacts and long-term skill development concerns, helping students recognize problematic patterns before significant academic consequences develop.

Support system development should include: (1) screening procedures to identify at-risk students early in their academic careers; (2) counseling services trained in technology dependency issues; (3) peer support programs that help students develop healthy usage norms and (4) ongoing monitoring systems that track the effectiveness of intervention efforts and adjust approaches based on outcomes.

5.6. Limitations and future research directions

This study has several limitations that should inform interpretation and future research. Most significantly, the cross-sectional design limits causal inferences regarding the direction of relationships between dependency and academic performance. While our analysis controlled for prior academic performance and multiple regression indicated dependency predicted performance beyond selection effects, longitudinal studies tracking students before and after AI exposure would provide stronger evidence for causal relationships.

The cross-sectional limitation is particularly important given the potential for bidirectional relationships between dependency and academic performance. While our theoretical framework and statistical controls support the interpretation that dependency leads to performance decline, it is also possible that students with academic difficulties are more likely to develop dependency patterns as they seek technological solutions to their struggles. Resolving this causal ambiguity requires longitudinal research designs that can establish temporal precedence between variables.

We propose a comprehensive three-wave longitudinal study tracking students from AI introduction through graduation to examine changes in dependency patterns and academic impacts over time. This design would include: (1) pre-AI baseline measurements of academic performance and relevant psychological variables; (2) regular assessment of AI usage patterns, dependency development and academic outcomes throughout the study period; (3) post-graduation follow-up to examine long-term consequences of different AI usage patterns and (4) experimental components that randomly assign students to different AI integration approaches to establish causal relationships.

The rapid evolution of generative AI capabilities means findings may not generalize to future iterations of these technologies. As AI systems continue to advance, usage patterns and dependency manifestations may shift, requiring ongoing investigation. Our focus on a single institution also limits generalizability across Zimbabwe's diverse higher education landscape, suggesting the need for multi-institutional studies capturing different resource levels and institutional cultures.

The technological evolution limitation requires adaptive research approaches that can track changing relationships between AI capabilities and user responses over time. Future research should employ flexible methodological frameworks that can accommodate rapid technological change while maintaining theoretical coherence and practical relevance.

Future research should extend this investigation to multiple institutions across different resource levels and cultural backgrounds to establish the generalizability of findings and identify culture-specific factors that influence dependency development. Comparative studies between developing and developed nation settings would be particularly valuable for understanding how environmental factors shape dependency patterns and intervention effectiveness.

Additionally, discipline-specific investigations could reveal how different academic domains create varying vulnerability to AI dependency. Fields that rely heavily on writing, critical analysis or creative problem-solving may show different patterns compared to those emphasizing factual knowledge or technical skills. Such research would inform the development of discipline-specific intervention approaches.

5.6.1. Intervention development and evaluation

Perhaps, most importantly, future research should focus on developing and evaluating intervention approaches specifically tailored to generative AI dependency in educational settings. Such research should examine: (1) the effectiveness of individual versus institutional intervention approaches; (2) the optimal timing for intervention efforts; (3) the role of peer support and social norms in maintaining behavior change and (4) the long-term sustainability of different intervention strategies.

The gap in intervention research represents a critical limitation in the current literature and a priority area for future investigation. As AI dependency becomes more prevalent and its consequences become clearer, evidence-based intervention approaches will become essential for protecting student welfare and educational outcomes.

6. Conclusion

This study demonstrates that generative AI dependency represents an emerging academic crisis among Zimbabwean university students, with significant implications for educational outcomes and skill development. The clear relationship between dependency severity and academic performance underscores the importance of addressing this issue within higher education environments, particularly in circumstances where generative AI serves dual roles as both a dependency vector and an essential educational resource substitution.

The identified contextual factors—infrastructure limitations, economic pressures, institutional resource constraints, policy ambiguity and career preparation rationalization—illustrate how dependency development in this environment represents a complex interaction between universal psychological mechanisms and specific environmental conditions. This suggests the need for contextualized approaches to both understanding and addressing generative AI dependency in developing nation university settings.

Three main pathways mediate between dependency and academic performance: critical thinking atrophy, writing skill degradation and knowledge acquisition reduction. These are general targets for the development of interventions. Examining these mechanisms, we can minimize the amount of data required for large-scale models to be accurate for a wider range of cases. Educational stakeholders may be better able to mitigate academic impacts, even amid dependency patterns. These findings hold crucial implications for higher education institutions in Zimbabwe in terms of crafting nuanced responses that recognize the promise of generative AI in resource-challenged settings and its potential to compromise basic educational redundancy. Instead of maintaining prohibition-focused policies bound to fail on the ground, institutions could take lessons from ecology, attending to the contextual factors identified, but also helping students build metacognitive skills necessary for the wise application of technology.

As generative AI continues its rapid evolution and integration into academic settings worldwide, understanding the specific dependency vulnerabilities and consequences in diverse settings becomes increasingly important. This study contributes to that understanding by examining an emerging technological dependency within Zimbabwe's distinctive higher education landscape, offering insights relevant to both dependency theory and educational practice in developing nations.

Future research should address these limitations through longitudinal designs that clarify causality, examination of intervention effectiveness and broader sampling across multiple institutions. Particular attention should be paid to developing valid assessment methods for generative AI dependency as the technology continues to evolve, and to understanding the long-term developmental impacts of cognitive skill displacement during university years.

Additionally, future studies should investigate potential protective factors that enable some students to maintain healthy AI usage patterns despite similar contextual challenges. Identifying both individual and environmental characteristics that promote resilience could inform more effective intervention approaches. Research exploring the transferability of our findings to other resource-constrained educational contexts would also contribute valuable comparative insights.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

All the data presented in this study can be made available upon request.

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